

# CALCULUS (II)

## WEEK (II)

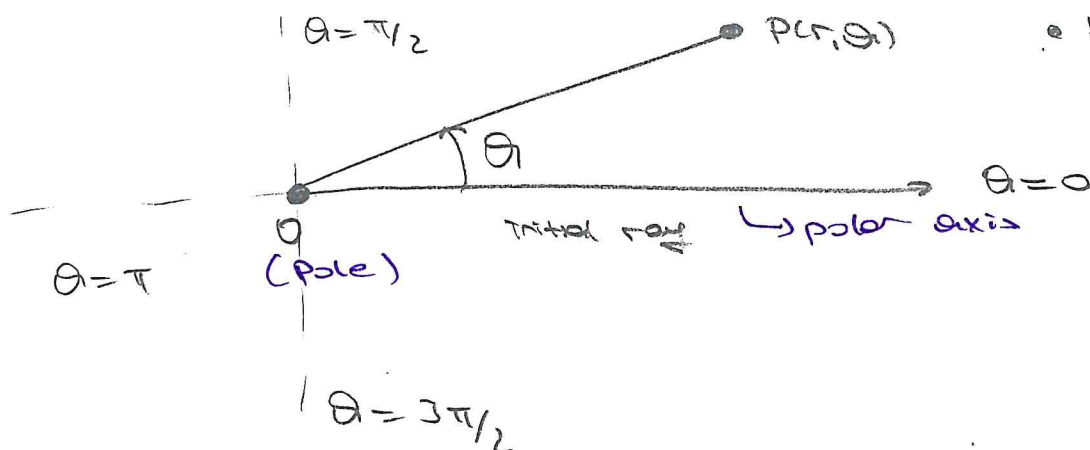
### POLAR COORDINATES AND POLAR CURVES

→ Usually we use cartesian coordinates, which are directed distances from two perpendicular (rectangular) axes. Here, we describe a new coordinate system called the polar coordinate system.

- Choose a point in the plane called **pole**  $\circ$  (origin) (cutup)

- Draw a ray (line) starting at  $\circ$  called the polar axis.

This axis is usually drawn horizontally to the right  
Corresponds to the positive x-axis in cartesian coordinates



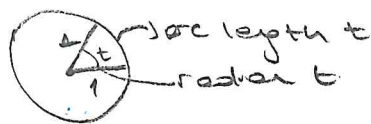
- P is any other point in the plane

- $r = |OP|$

- $\theta$ : angle

(usually measured in radians  $2\pi \text{ rad} = 360^\circ$ )

Radian



$\pi \text{ radians} = 180^\circ$

of P.

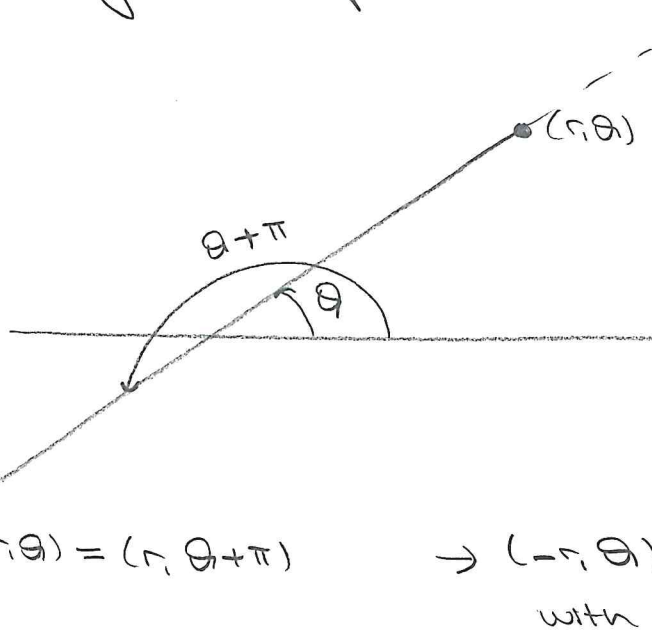
- the ordered pair  $(r, \theta)$  and  $r, \theta$  are called polar coordinate

→ Here, we will use the convention (or idea) that an angle

is: **positive / negative** if measured in the **counterclockwise /**

**clockwise** direction

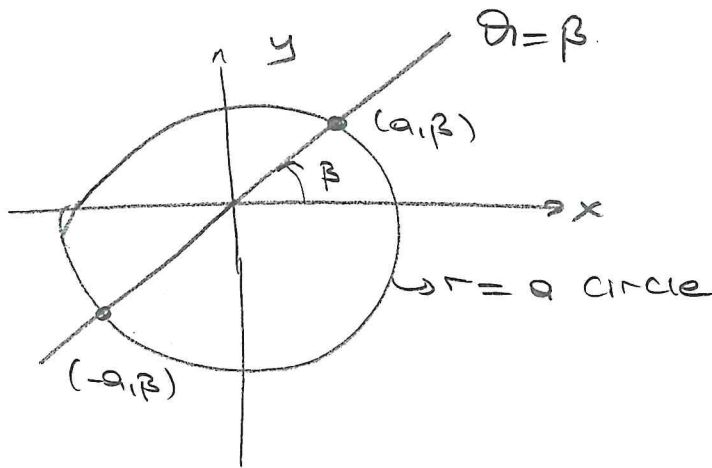
- Polar coordinates are not unique.
- Polar coordinates can have negative  $r$  values
- If  $\theta = 0$ , then  $r=0$  and  $(0, \theta)$  represents the pole for any value of  $\theta$



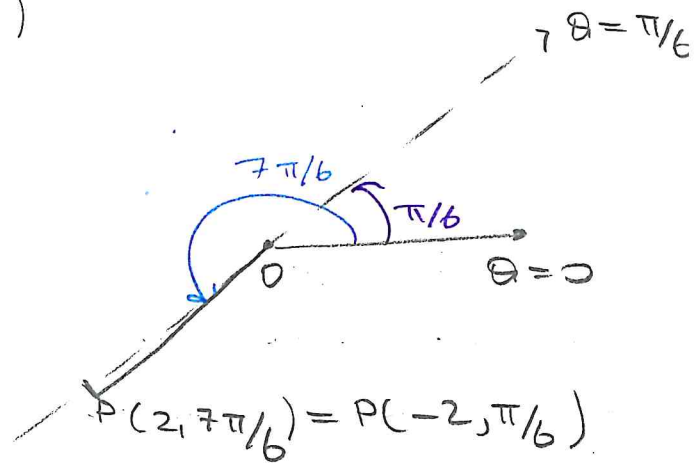
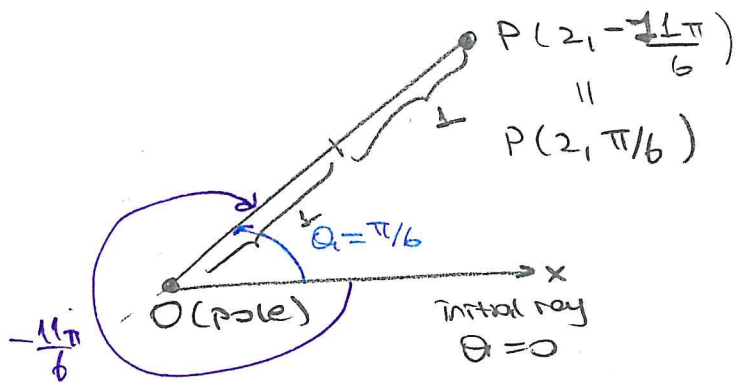
$\rightarrow \theta \rightarrow$  The point  $(r, \theta)$  and  $(-r, \theta)$  on the same line through  $O$  and at the same distance  $(|r|)$  from  $O$ , but opposite sides of  $O$ .

$\rightarrow (-r, \theta)$  represents the same point with  $(r, \theta + \pi)$

NOT



- $\rightarrow$  As in trigonometry,  $\theta$  is positive when measured counterclockwise and negative when measured clockwise
- $\rightarrow$  The angle associated with a given point is not unique

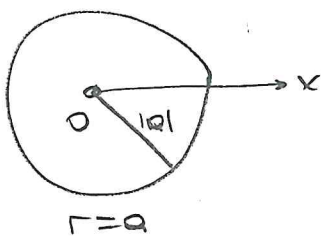


$P(2, 7\pi/6)$ :  $\theta = 7\pi/6$ , go 2 steps forward.  
 $P(-2, \pi/6)$ :  $\theta = \pi/6$ , go 2 steps backward.

### Equation

$$r = a$$

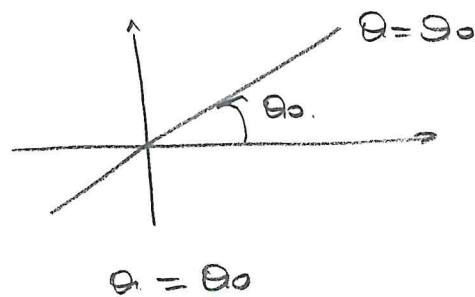
$$\theta = \theta_0$$



### Graph

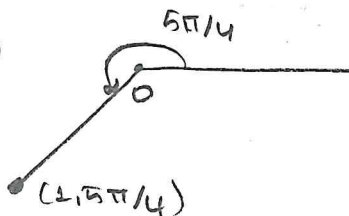
Circle of radius  $|a|$  centered at  $O$

Line through  $O$  making an angle  $\theta_0$  with the initial ray

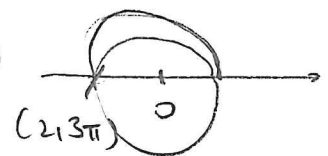


Ex 7 Plot the following points

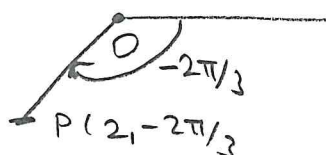
(a)  $(1, 5\pi/4)$



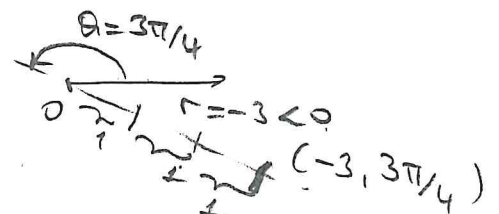
(b)  $(2, 3\pi)$



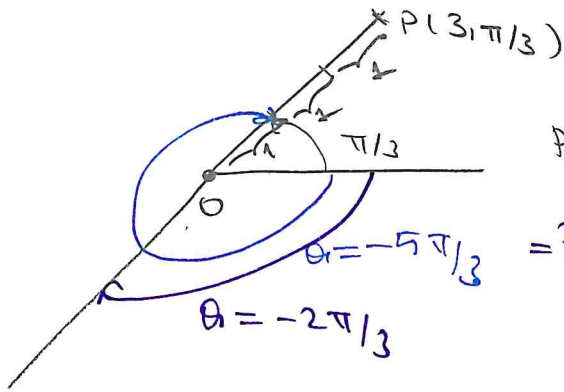
(c)  $(2, -2\pi/3)$



(d)



Ex Find all the polar coordinates from  $P(3, \pi/3)$



$$\begin{aligned} P(3, \pi/3) &= P(3, -5\pi/3) = \\ &= P(-3, -2\pi/3) = P(-3, \underbrace{4\pi/3}_{-2\pi/3 + 2\pi}) \end{aligned}$$

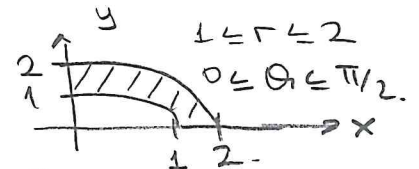
So, we have for

$$r = 3 \quad \pi/3, \pi/3 + 2\pi, \pi/3 + 4\pi, \dots \quad (3, \pi/3 + 2n\pi) \quad n=0, \pm 1, \pm 2, \dots$$

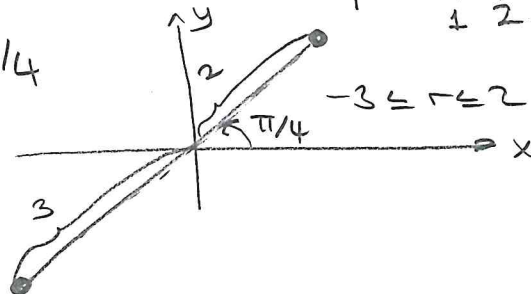
$$r = -3 \quad -2\pi/3, -2\pi/3 + 2\pi, -2\pi/3 + 4\pi, \dots \quad (-3, -2\pi/3 + 2n\pi) \quad n=0, \pm 1, \pm 2, \dots$$

Ex Graph the sets of points whose polar coordinates satisfy the following conditions

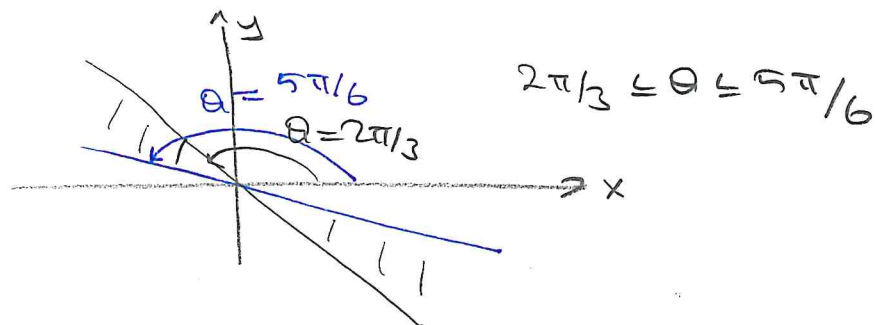
①  $1 \leq r \leq 2$  and  $0 \leq \theta \leq \pi/2$



②  $-3 \leq r \leq 2$  and  $\theta = \pi/4$

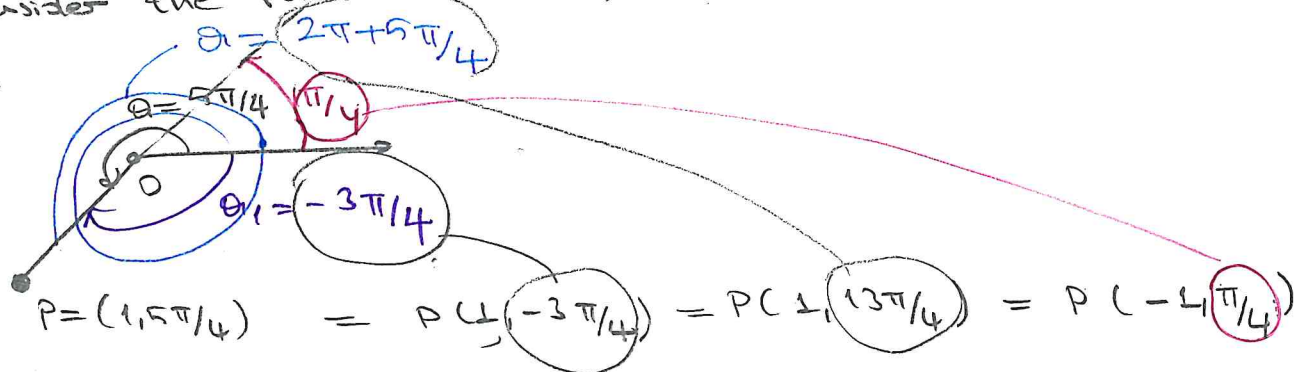


③  $\frac{2\pi}{3} \leq \theta \leq \frac{5\pi}{6}$  (no restriction on  $r$ )



NOTE In cartesian coordinate system every point has a only one (unique) representation but in the polar coordinate each point has many representation

Ex Consider the point  $(1, 5\pi/4)$ . Represent this point in different ways.



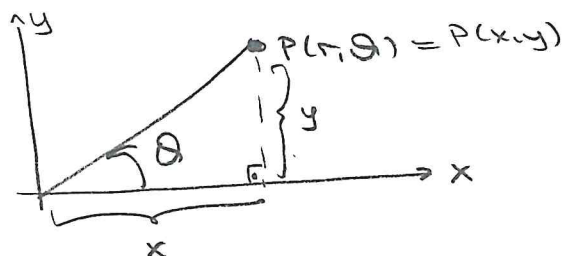
NOTE Since a complete counterclockwise rotation is given by an angle  $2\pi$ , the point  $(r, \theta)$  is also represented by

\*\*\*  $(r, \theta + 2n\pi)$  or  $(-r, \theta + (2n+1)\pi)$   $n$  is any integer

## POLAR COORDINATES AND GRAPHS

### CONNECTION BETWEEN POLAR AND CARTESIAN COORDINATES

→ The pole coordinates to the origin and the polar axis coincides with the positive  $x$ -axis



$$\cos \theta = \frac{x}{r} \quad \sin \theta = \frac{y}{r}$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$\tan \theta = \frac{y}{x}$$

$$\theta = \arctan\left(\frac{y}{x}\right)$$

$$-\pi/2 < \theta < \pi/2$$

NOTE Although these equations are derived for  $r > 0$  and  $0 < \theta < \pi/2$ , they are valid for all values of  $r$  and  $\theta$ . From these equations we can obtain that

$$x^2 + y^2 = r^2$$



Ex 7 Here are some equivalent equations expressed in terms of both polar coordinate and Cartesian coordinates

### Polar Equation

$$r \cos \theta = 2$$

$$r^2 \cos \theta \sin \theta = 4$$

$$r^2 \cos^2 \theta - r^2 \sin^2 \theta = 1$$

$$r = 1 + 2r \cos \theta$$

$$r = 1 - \cos \theta$$

### Cartesian Equation

$$x = 2$$

$$xy = 4$$

$$x^2 - y^2 = 1$$

$$y^2 - 3x^2 - 4x - 1 = 0$$

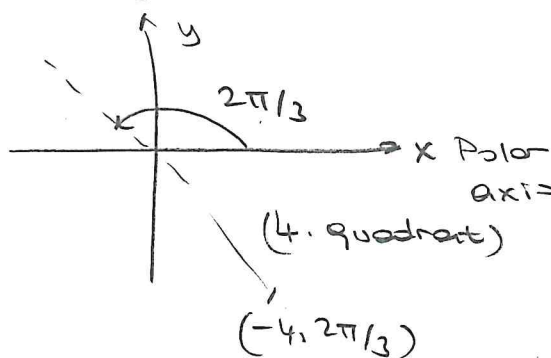
$$\begin{cases} r^2 = (1 + 2r \cos \theta)^2 \\ x^2 + y^2 = (1 + 2x)^2 \\ x^2 + y^2 = 1 + 4x + 4x^2 \end{cases}$$

$$x^4 + y^4 + 2x^2y^2 + 2x^3 + 2xy^2 - y^2 = 0$$

$$\begin{cases} (r + \cos \theta)^2 = 1 \\ r^2 + 2r \cos \theta + \cos^2 \theta = 1 \end{cases}$$

Ex 7 Convert each of the following points into the given coordinate system.

(a)  $(-4, 2\pi/3)$  into cartesian coordinate



$$r = -4 \quad \theta = 2\pi/3$$

$$x = r \cos \theta = -4 \cdot \cos(2\pi/3) = -4(-1/2) = 2$$

$$y = r \sin \theta = -4 \sin(2\pi/3) = -4 \cdot \frac{\sqrt{3}}{2} = -2\sqrt{3}$$

$$\text{Polar } (-4, 2\pi/3)$$

$$\text{Cartesian } (2, -2\sqrt{3})$$

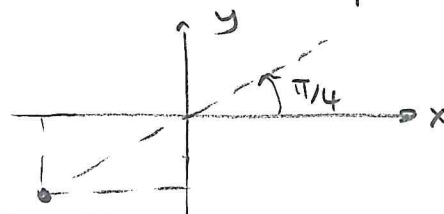
(b)  $(-1, -1)$  into polar coordinates.

$$x = -1 \quad y = -1 \quad \tan \theta = \frac{-1}{-1} = 1 \Rightarrow \theta = \arctan(1) = \pi/4$$

But this is NOT THE CORRECT ANSWER. Since this value of  $\theta$  is in the first quadrant but the point is in the 3rd quadrant.

$$\theta = \pi/4 + \pi = 5\pi/4 \quad r = \sqrt{1+1} = \sqrt{2}$$

Therefore, the point is  $(\sqrt{2}, 5\pi/4)$  in Polar coordinates



Ex 7. (a) Convert  $2x - 7x^3 = 1 + xy$  into polar coordinates.

Substitute  $x = r \cos \theta$   $y = r \sin \theta$

$$2r \cos \theta - 7r^3 \sin^3 \theta = 1 + r^2 \sin \theta \cos \theta$$

(b) Convert  $r = -b \cos \theta$  into cartesian coordinates

Recall  $r^2 = x^2 + y^2$   $x = r \cos \theta$   $y = r \sin \theta$

$$r = -b \cos \theta \Rightarrow r^2 = -b r \cos \theta$$

multiply  
with  $r$   $x^2 + y^2 = -bx$

Ex 7 Find a polar equation for the circle  $x^2 + (y-3)^2 = 9$

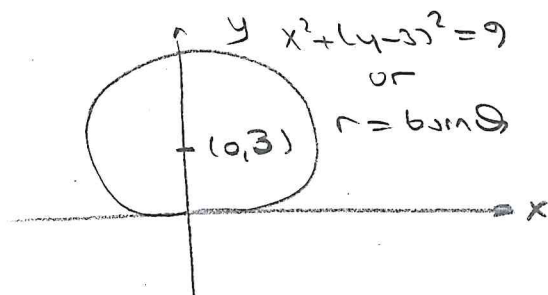
$$x^2 + (y-3)^2 = 9 \Rightarrow x^2 + y^2 - 6y + 9 = 9$$

$$r^2 - 6r \sin \theta = 0$$

$$r(r - 6 \sin \theta) = 0$$

$$r = 0 \text{ or } r - 6 \sin \theta = 0$$

$$r = 6 \sin \theta$$



## POLAR CURVES

Definition (Polar Curve) The graph of a polar equation  $F(r, \theta) = 0$  consists of all points  $P$  that have at least one polar representation  $(r, \theta)$  whose coordinates satisfy the equation.

Ex 1  $r = a$

No matter what angle, we have the distance from the origin must be  $|a|$ . This is exactly the definition of circle of radius  $|a|$  centered at the origin.

Ex 2  $\theta = \beta$  ( $\beta$ : constant in radians)

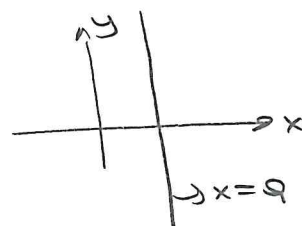
$$\theta = \arctan(y/x) = \beta \Rightarrow \tan(\beta) = y/x$$

$$y = (\tan \beta) x$$

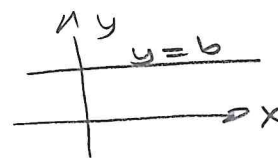
This is a line that goes through the origin and makes an angle of  $\beta$  with the positive x-axis



Ex 3  $\underbrace{r \cos \theta}_x = a \Rightarrow x = a$  Indeed, we have  $x = a$  which is a vertical line



Ex 4  $\underbrace{r \sin \theta}_y = b \Rightarrow y = b$  This converts  $y = b$  which is a horizontal line



Ex 5  $r = 2a \cos \theta$

$$r = x^2 + y^2 \quad \cos \theta = x/r \Rightarrow r = 2a \cos \theta$$

$$r = 2a \cdot \frac{x}{r} \Rightarrow r^2 = 2ax \quad x^2 + y^2 = 2ax$$

$$x^2 - 2ax + a^2 - a^2 + y^2 = 0 \Rightarrow (x-a)^2 + y^2 = a^2 \quad M(a,0) \text{ center} \quad r = |a|$$

Ex 6  $r = 2b \sin \theta$

$$\sin \theta = \frac{y}{r} \Rightarrow r = 2b \frac{y}{r} \Rightarrow r^2 = 2by$$

$$x^2 + y^2 = 2by \quad x^2 + y^2 - 2by + b^2 = b^2 \quad x^2 + (y-b)^2 = b^2$$

$M(0,b) \quad r = |b|$  circle



Ex 7  $r = 2a \cos \theta + 2b \sin \theta$

$\cos \theta = \frac{x}{r} \quad \sin \theta = \frac{y}{r} \Rightarrow r = 2a \frac{x}{r} + 2b \frac{y}{r}$

$r^2 = 2ax + 2by \quad x^2 + y^2 = 2ax + 2by$

$x^2 - 2ax + a^2 - a^2 + y^2 - 2by + b^2 - b^2 = 0$

$(x-a)^2 + (y-b)^2 = a^2 + b^2 \quad r = \sqrt{a^2 + b^2} \quad M(a,b) \text{ circle.}$

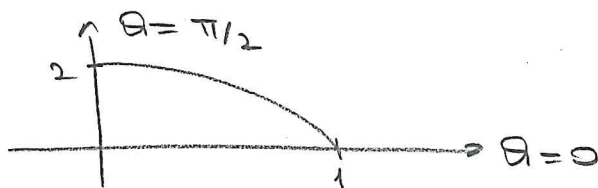
Ex 8 Graph  $r=7$ ,  $r=4 \cos \theta$  and  $r=7 \sin \theta$  on the same axis system (solve after definition)

## CARDIOIDS ( $r = a \mp a \cos \theta$ , $r = a \mp a \sin \theta$ )

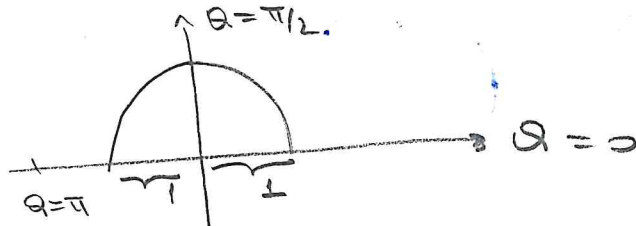
→ These have graph resembles to a heart and always contain the origin.

→ For instance consider  $r = 1 + \sin \theta$

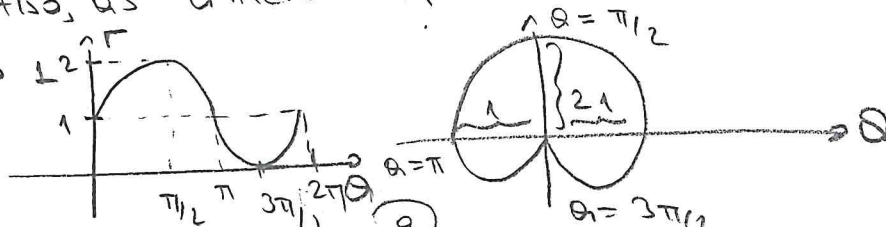
- As  $\theta$  increases from 0 to  $\pi/2$ ,  $r$  increases from 1 to 2



- As  $\theta$  increases from  $\theta = \pi/2$  to  $\pi$ ,  $r$  decreases from 2 to 1



- As  $\theta$  increases from  $\pi$  to  $3\pi/2$ ,  $r$  decreases from 1 to 0. Also, as  $\theta$  increases from  $3\pi/2$  to  $2\pi$ ,  $r$  increases from 0 to 1



## LIMACONS

### Limacons with an inner loop

$$r = a \mp b \cos \theta \text{ and } r = a \mp b \sin \theta \quad a < b$$

These have an inner loop and always contain the origin

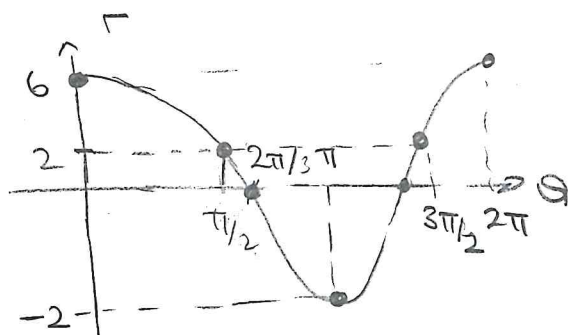
### Limacons without an inner loop

$$r = a \mp b \cos \theta \text{ and } r = a \mp b \sin \theta \quad a > b$$

These do not have an inner loop and do not contain the origin

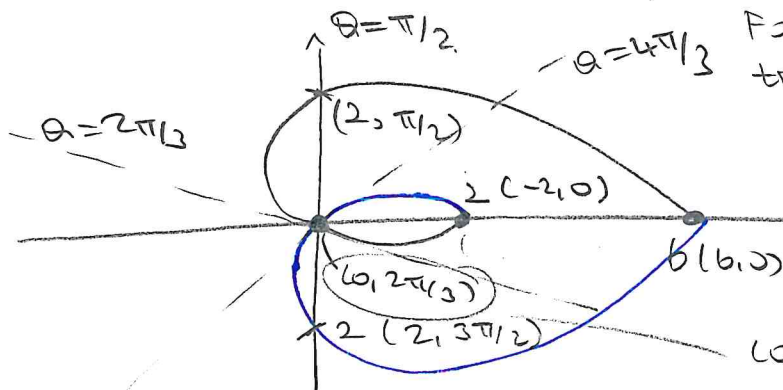
Ex  $r = 2 + 4 \cos \theta$   $a = 2$   $b = 4$   $a < b$

| <u>Points</u> | $r$ | $\theta$ |
|---------------|-----|----------|
|               | 6   | 0        |
|               | 2   | $\pi/2$  |
|               | 0   | $2\pi/3$ |
|               | -2  | $\pi$    |



$$\begin{aligned} \cos(2\pi/3) &= -\cos(\pi - \pi/3) \\ &= -\cos(\pi/3) = -1/2 \end{aligned}$$

For our graph, we will draw till  $\pi$  then take symmetry



Limaçon with an inner loop.

polar  $(0, 2\pi/3)$  corresponds to  $(0, 0)$  in cartesian.

$$\begin{aligned} x &= r \cos \theta & y &= r \sin \theta \\ x &= 0 \cdot \cos 2\pi/3 & y &= 0 \cdot \sin 2\pi/3 \\ &= 0 & &= 0 \end{aligned}$$

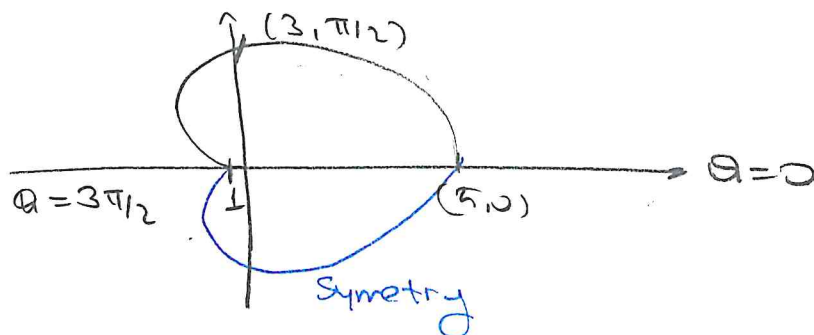
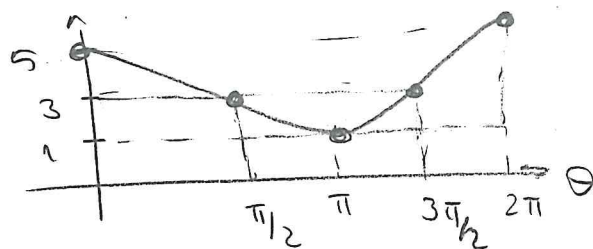
Since  $2 + 4 \cos \theta = 2 + 4 \cos(-\theta)$  this means that we have a symmetry with the polar axis. Symmetry is given by the blue part.

Ex Sketch the graph of  $r = 3 + 2\cos\theta$   $a=3$   $b=2$   
 $a > b$

Points

$r$   
 5  
 3  
 1  
 3  
 5

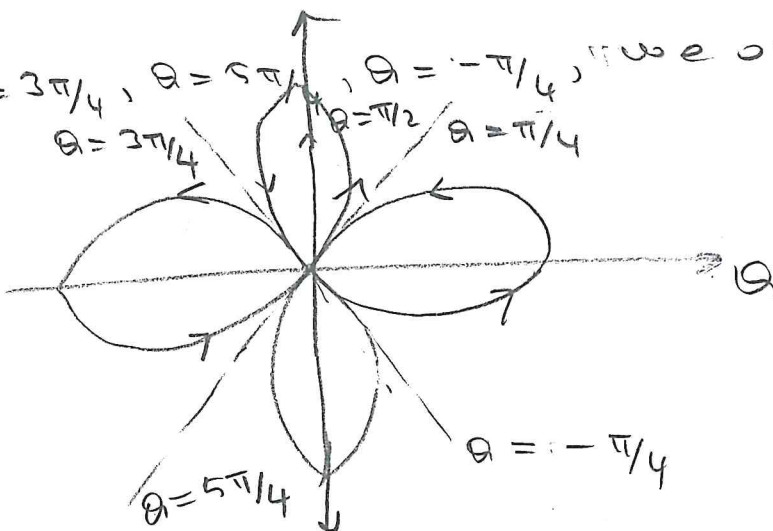
$\theta$   
 0  
 $\pi/2$   
 $\pi$   
 $3\pi/2$   
 $2\pi$



FOUR LEAVED ROSE

$$r = \cos 2\theta$$

For  $\theta = \pi/4, \theta = 3\pi/4, \theta = 5\pi/4, \theta = 7\pi/4$ , we obtain  $r = 0$



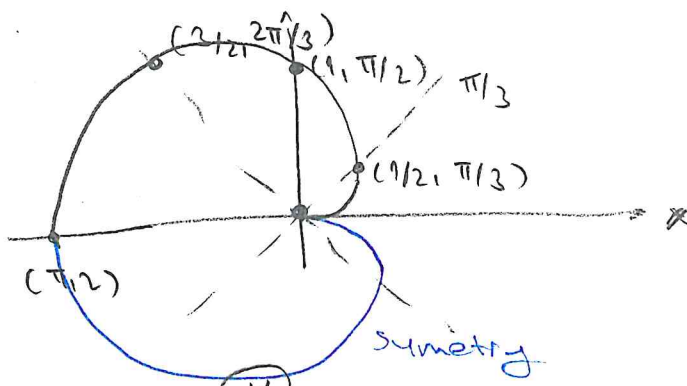
Ex Graph the curve  $r = 1 - \cos\theta$ .

$(r, \theta)$  on the graph  $\Rightarrow r = 1 - \cos\theta$

$$r = 1 - \cos(-\theta)$$

$(r, -\theta)$  on the graph

| $\theta$ | $r = 1 - \cos\theta$ |
|----------|----------------------|
| 0        | 0                    |
| $\pi/3$  | $1/2$                |
| $\pi/2$  | 1                    |
| $2\pi/3$ | $3/2$                |
| $\pi$    | 2                    |



Ex 7 Graph the curve  $r^2 = 4\cos\theta$

Solution The equation  $r^2 = 4\cos\theta$  requires  $\cos\theta \geq 0$ , so we get the entire graph by running  $\theta$  from  $-\pi/2$  to  $\pi/2$

The curve is symmetric about the x-axis because

$$(r, \theta) \text{ on the graph} \Rightarrow r^2 = 4\cos\theta$$

$$r^2 = 4\cos(-\theta)$$

$$(r, -\theta) \text{ on the graph}$$

The curve is also symmetric about the origin because

$$(r, \theta) \text{ on the graph} \Rightarrow r^2 = 4\cos\theta$$

$$(-r)^2 = 4\cos\theta$$

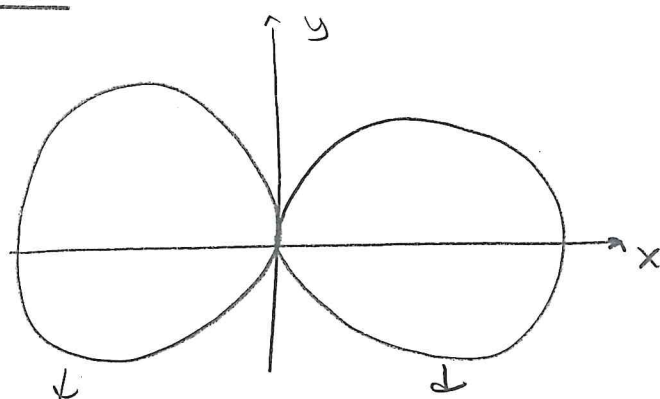
$$(-r, \theta) \text{ on the graph}$$

The curve passes through the origin when  $\theta = -\pi/2$  and  $\theta = \pi/2$ . It has a vertical tangent both times because  $\tan\theta$  is infinite

For each value of  $\theta$  in the interval between  $-\pi/2$  and  $\pi/2$ , the formula  $r^2 = 4\cos\theta$  gives two values of  $r$

$$r = \pm 2\sqrt{\cos\theta}$$

| $\theta$   | $\cos\theta$         | $r = \pm 2\sqrt{\cos\theta}$ |
|------------|----------------------|------------------------------|
| 0          | 1                    | $\pm 2$                      |
| $\pm\pi/6$ | $\frac{\sqrt{3}}{2}$ | $\approx \pm 1.9$            |
| $\pm\pi/4$ | $\frac{1}{\sqrt{2}}$ | $\approx \pm 1.7$            |
| $\pm\pi/3$ | $\frac{1}{2}$        | $\approx \pm 1.4$            |
| $\pm\pi/2$ | 0                    | 0                            |



loop for

$$r = -2\sqrt{\cos\theta}$$

$$-\pi/2 \leq \theta \leq \pi/2$$

loop for

$$r = 2\sqrt{\cos\theta}$$

$$-\pi/2 \leq \theta \leq \pi/2$$

$$r^2 = 4\cos\theta \quad x^2 + y^2 = 4 \cdot \frac{x}{r}$$

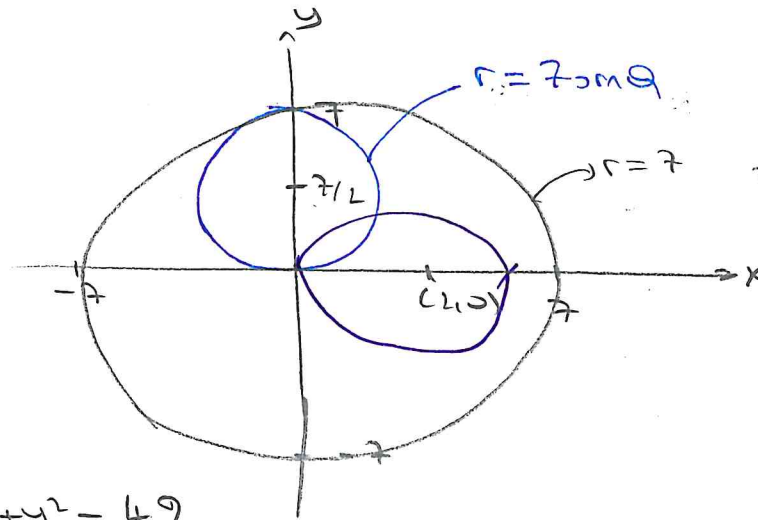
$$(x^2 + y^2)^{3/2} = 4 \cdot x$$

Ex 7 Graph  $r=7$ ,  $r=4\cos\theta$  and  $r=7\sin\theta$  on the same axis system

$$r = \sqrt{x^2 + y^2}$$

$$\cos\theta = \frac{x}{r}$$

$$\sin\theta = \frac{y}{r}$$



$$r = 7 \quad x^2 + y^2 = 49$$

$$r = 4\cos\theta \Rightarrow r = 4 \frac{x}{r} \Rightarrow r^2 = 4x \quad x^2 + y^2 = 4x$$

$$(x-2)^2 + y^2 = 4$$

$$M(2,0) \quad r=2$$

$$r = 7\sin\theta \Rightarrow r = 7 \frac{y}{r} \quad r^2 = 7y \quad x^2 + y^2 = 7y$$

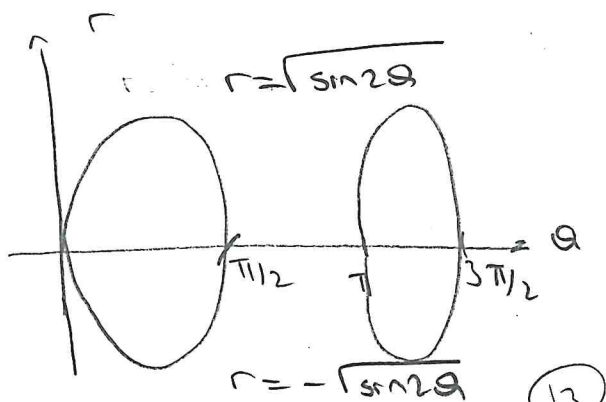
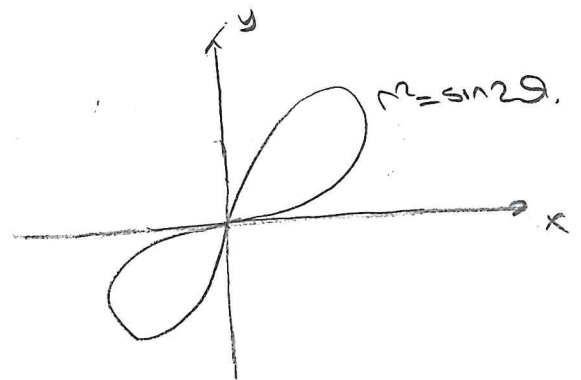
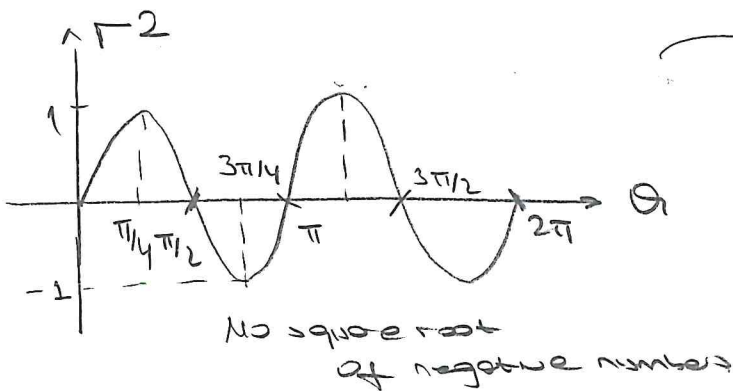
$$x^2 + y^2 - 7y = 0$$

$$x^2 + (y - 7/2)^2 = 49/4$$

$$M(0, 7/2) \quad r=7/2$$

Ex 7 Graph the lemniscate curve

$$r^2 = \sin 2\theta$$

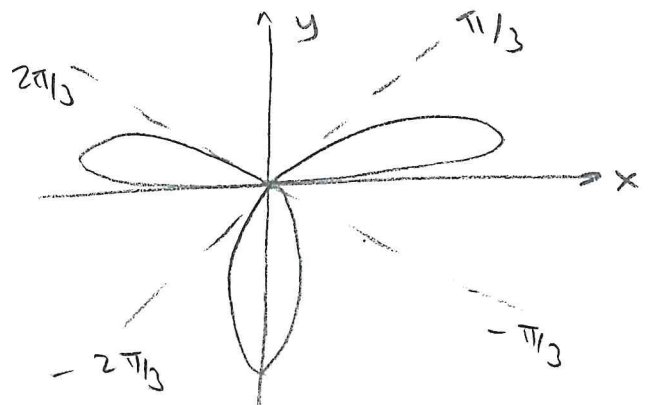
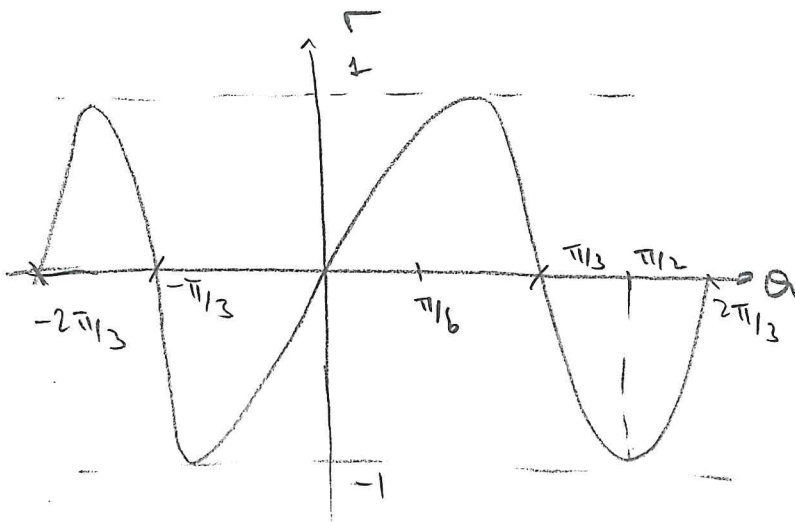
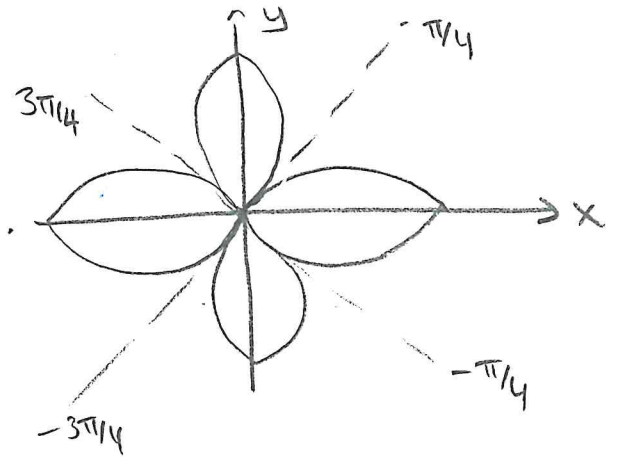
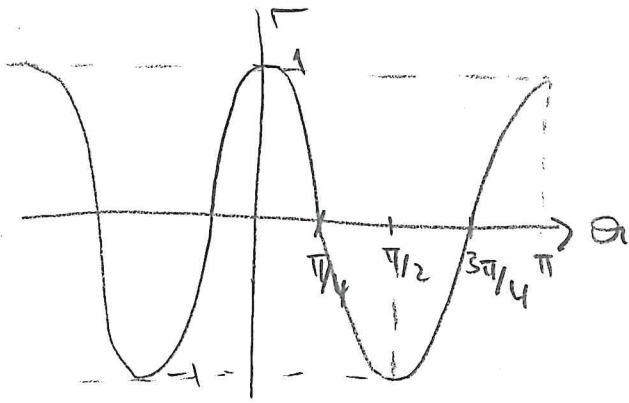




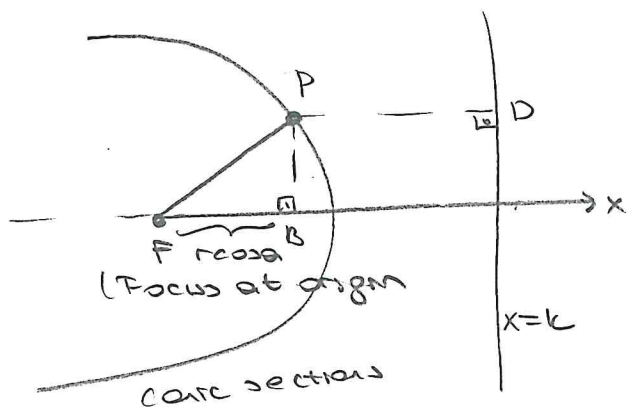
Ex Sketch the polar graphs

②  $r = \cos 2\theta$

③  $r = \sin 3\theta$



# CONICS IN POLAR COORDINATES (NOT GIVEN TO STUDENTS) Pages 15-17



$$|PF| = r$$

$$|PD| = k - FB = k - r \cos \theta$$

$$PF = e \cdot PD$$

(Eccentricity)

distance of P from the

$$e = \frac{\text{distance of P from the origin}}{\text{perpendicular distance from P to D}}$$

$$|PF| = e |PD|$$

$$r = e (k - r \cos \theta)$$

Polar Equation for a conic with Eccentricity e

$$r = \frac{ke}{1 + e \cos \theta} \quad \text{where } x=k > 0 \text{ is the vertical directrix}$$

Ex Here are polar equations for three conics. The eccentricity values identifying the conic are the same for both polar and Cartesian coordinates.

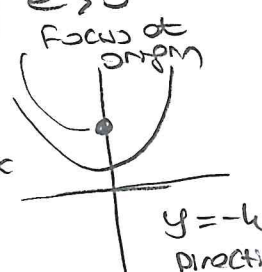
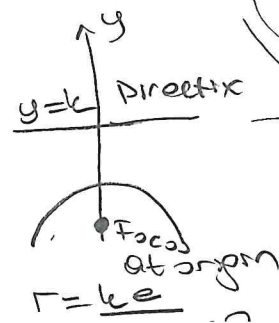
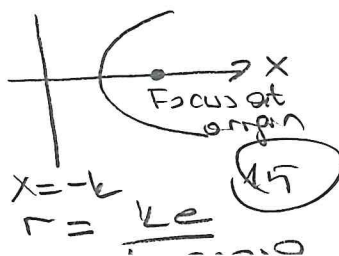
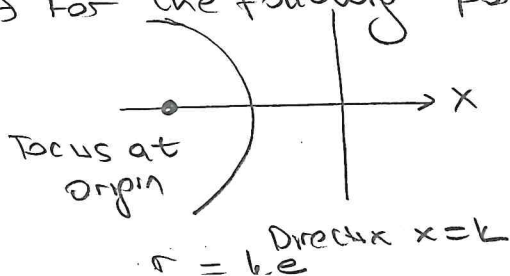
$$e = 1/2 \quad r = \frac{k}{2 + \cos \theta} \quad \text{ellipse}$$

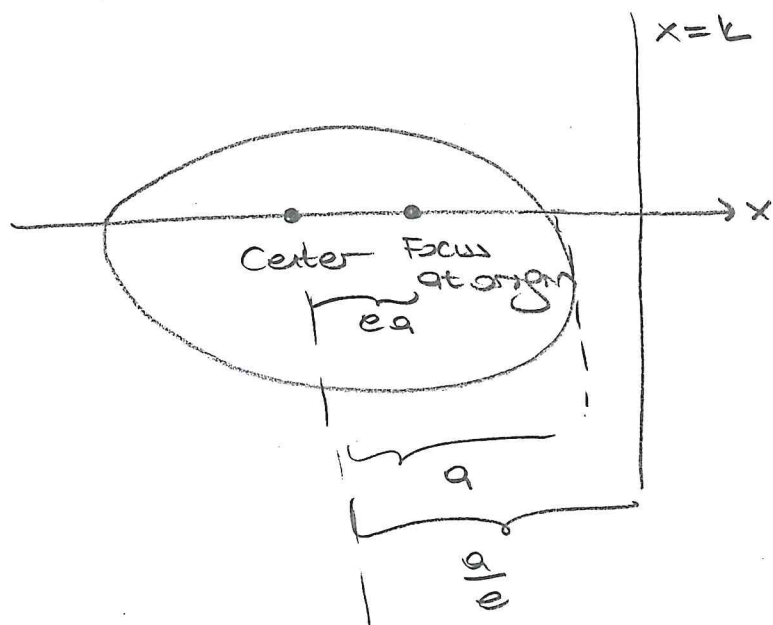
$$e = 1 \quad r = \frac{k}{1 + \cos \theta} \quad \text{parabola}$$

$$e = 2 \quad r = \frac{2k}{1 + 2 \cos \theta} \quad \text{hyperbola}$$

→ Equations for conic sections with eccentricity  $e > 0$  but different locations of the directrix

→ For the following parabolas  $e = 1$ .

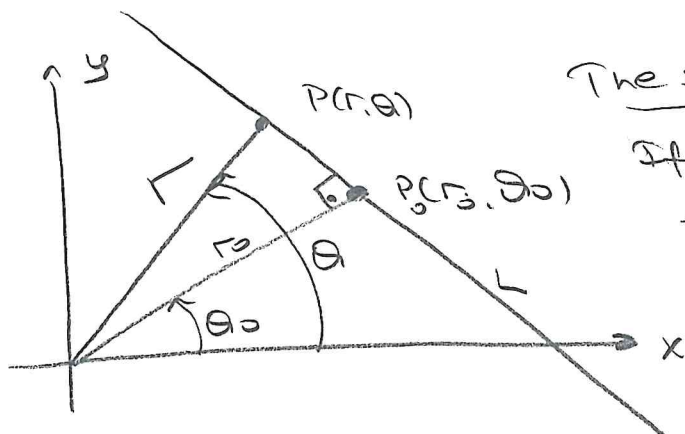




In an ellipse with  
Semi-major axis  $a$ ,  
the focus-directrix  
distance is  
 $k = (a/e) - ea > 0$   
 $ke = a(1-e^2)$

Polar Equation for the Ellipse  
with Eccentricity  $e$  and  
Semi-major Axis  $a$

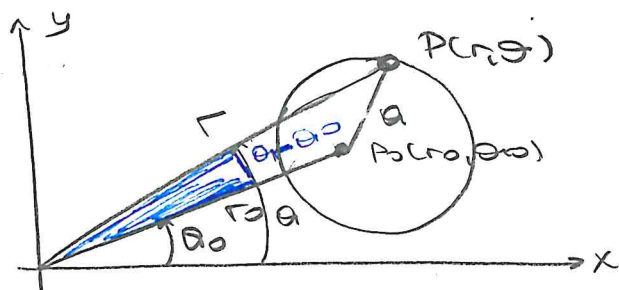
$$r = \frac{a(1-e^2)}{1+e \cos \theta}$$



The standard Polar Equation for Line

If the point  $P_0(r_0, \theta_0)$  is  
the foot of the perpendicular  
from the origin to the line  
 $L$ , and  $r_0 > 0$ , then an equation  
for  $L$  is

$$r \cos(\theta - \theta_0) = r_0$$



$$a^2 = r^2 + r_0^2 - 2rr_0 \cos(\theta - \theta_0)$$

Law of cosine

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Note that equation  $a^2 = r^2 + r_0^2 - 2rr_0 \cos(\theta - \theta_0)$  covers the case when the points O, P and P\_0 doesn't give a triangle (i.e., when  $\theta = \theta_0$ )

→ If circle passes through origin  $r_0 = a$  and we get

$$a^2 = a^2 + r^2 - 2ar \cos(\theta - \theta_0)$$

$$r^2 = 2ar \cos(\theta - \theta_0)$$

If circle's center lies on the positive x axis, i.e.  $\theta_0 = 0$ ,

$$r = 2a \cos \theta$$

→ If center lies on positive y-axis  $\theta = \pi/2$   $\cos(\theta - \pi/2) = \sin$

So, the equation  $r = 2a \cos(\theta - \theta_0)$  becomes

$$r = 2a \sin \theta$$

→ Equations for circles through the origin centered on negative x- and y-axes can be replaced by replacing  $r$  with  $-r$  so we get  $r = -2a \cos \theta$

$$r = -2a \sin \theta$$

